Row-Level Structural Fidelity Diagnostics for Synthetic Tabular Data

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Abstract

Evaluating synthetic tabular data requires auditing individual record integrity in addition to global distribution alignment. Standard statistical metrics (KS complement, Total Variation Distance, Correlation Similarity) assess column distributions in aggregate and are fundamentally blind to localized, row-level conditional dependency violations. We present the Hybrid Integrity Framework (HIF), a row-level structural diagnostic that combines a Logical Sentinel Ensemble (LSE) for categorical conditional dependencies with Neighbor-Invariant Continuity (NIC) for continuous feature consistency. We demonstrate two primary contributions: (1) **Empirically Robust Structural Diagnostics**: across tested generators (Gaussian Copula, Vine Copula, CTGAN, TVAE) and five benchmark datasets (Adult, Credit Default, Census ACS, Online Purchases, Supermarket Sales), HIF identifies row-level dependency ruptures even when aggregate marginal metrics report near-perfect fidelity (KS up to $\approx \mathbf{98\%}$); and (2) **Targeted Remediation**: pruning dependency-violating records ($\boldsymbol{H < 0.5}$) improves downstream predictive utility when structural errors are present (e.g., **+12.5** points F1 on Census ACS CTGAN, paired $\boldsymbol{t}$-test $\boldsymbol{p = 0.002}$; **+22.6** points on Online Purchases CTGAN) while preserving predictive utility in settings where structural errors are weak or absent (Gaussian Copula, paired $\boldsymbol{t}$-test $\boldsymbol{p = 0.13}$). HIF provides practitioners with a practical row-level signal of joint structural integrity for high-stakes synthetic data deployment.

1 Introduction

The generation of high-fidelity synthetic tabular data has emerged as a cornerstone of reproducible research in privacy-sensitive domains such as economics [1], finance, and healthcare [2, 3]. With the scaling of deep generative architectures, such as CTGAN [4] and the refinement of flexible statistical models, such as Vine Copulas [5], the community has achieved remarkable success in matching aggregate marginal distributions. However, a critical "Dependency Fidelity Gap" remains: while these models excel at replicating per-column distributions, they frequently produce records that are individually plausible under each column's marginal yet violate the conditional inter-feature dependencies of the real-world joint distribution [6]—violations invisible to any metric that evaluates columns in isolation.

Traditional evaluation pipelines rely on *statistical fidelity* metrics, such as the Kolmogorov-Smirnov test and Total Variation Distance, to assess quality [7, 8]. While useful for verifying marginal distributional alignment, these metrics are inherently *aggregation-blind*: they evaluate each column's distribution independently and cannot detect records that violate conditional inter-feature dependencies yet remain within the expected marginal range of each individual column. Recent attempts to bridge this gap, such as the SHAP-based semantic fidelity metric [9], provide explainability-aware insights but often lack the scalability and technical rigor required for the release of high-stakes synthetic data. As noted by Shang et al. (2026) [10], ensuring the integrity of synthetic tabular data requires more than marginal distributional similarity; it requires a *row-level dependency integrity certificate*. Crucially, this pathology is not exclusive to the opacity of deep neural networks: as we demonstrate, even rigorous statistical models (such as Vine Copulas) confidently hallucinate these joint dependency violations despite strong aggregate marginal fits.

To address these challenges, we introduce the Hybrid Integrity Framework (HIF), a system for detecting logical inconsistencies in synthetic tabular data. HIF reframes the problem of fidelity evaluation as a multidimensional consistency detection task. By moving beyond aggregate statistical snapshots, the HIF identifies logical violations at the record level, enabling the filtering and curation of high-integrity synthetic cohorts. In Section 2, we contextualize our approach within the emerging literature on semantic fidelity and inter-column logic before detailing our methodology in Algorithms 1 and 2.

Our framework comprises three complementary auditing layers:

1. **Logical Sentinel Ensemble (LSE):** An array of discriminative classifiers that identify high-dependency categorical feature hubs in the ground-truth data using cross-validated classification accuracy.
2. **Neighbor-Invariant Continuity (NIC):** A continuous feature auditing mechanism that verifies local semantic adjacency within SVD-embedded categorical contexts using robust percentile thresholding.
3. **Automated Rule Extractor:** An association rule mining engine that extracts high-confidence deterministic implication rules ($A \implies B$) from ground-truth data to enforce hard structural constraints.

Furthermore, we integrate the Targeted Adversarial Masking and Inference Suite (TAMIS) Privacy Oracle, subjecting synthetic cohorts to rigorous membership [11] and linkability attacks to ensure that logical consistency does not come at the cost of privacy.

Summary of Technical Novelty and Core Contributions.

We explicitly demarcate our contributions into two core pillars:

1. **Row-Level Structural Diagnostic (Evaluated Across Diverse Settings):** HIF reframes synthetic data auditing from aggregate column matching to record-level conditional plausibility. By combining predictive synergy hub selection with non-compensatory multiplicative aggregation ($\prod_l (1 - \mathcal{P}_l)$), HIF detects localized structural hallucinations ("impossible junctions") where every individual column value appears marginal-plausible yet their joint combination violates conditional logic.
2. **Targeted Remediation (Characterizable Utility Recovery):** We formalize and empirically characterize when integrity filtering improves downstream predictive utility. When synthetic cohorts contain structural errors (e.g., CTGAN on Census ACS), pruning low-HIF records ($H < 0.5$) yields statistically significant downstream utility gains (+12.5 points F1, paired t-test $p = 0.002$; +22.6 points on Online Purchases with CTGAN). When synthetic cohorts are largely free of structural errors (e.g., Gaussian Copula on Census ACS), HIF filtering preserves predictive utility without detectable degradation (paired t-test $p = 0.13$).

By providing a scalable, data-driven diagnostic, HIF enables the production of synthetic data certified for high-stakes research. The complete open-source implementation, experimental benchmarks, and dataset configurations are available at https://anonymous.4open.science/r/tabular-polygraph-21CE.

2 Related Work

Synthetic Tabular Data Generation.

The field has evolved from early copula-based methods to deep generative architectures. GAN-based models, such as CTGAN and TVAE [4], set an early benchmark for capturing joint distributions by building on the Synthetic Data Vault framework [12]. Recently, models such as CTAB-GAN [13] have improved the handling of skewed distributions. Concurrently, diffusion paradigms such as TabDDPM [14], flow-based gradient-boosted tree generators [15], and composable architectures such as CoDi [16] have pushed the frontier of manifold matching, while permutation-aware and federated objectives extend synthesis to decentralized tabular settings [17]. However, both statistical copula decompositions and deep neural architectures suffer from a fundamental objective mismatch: statistical models (such as Vine Copulas) excel at per-column marginal fitting but smooth over complex non-linear conditional constraints, while neural models (such as CTGAN) can learn non-linear categorical contexts but remain prone to mode collapse or localized joint violations. Our work focuses on a representative spectrum of generators, from classical *Gaussian Copulas* [18] and *Vine Copulas*

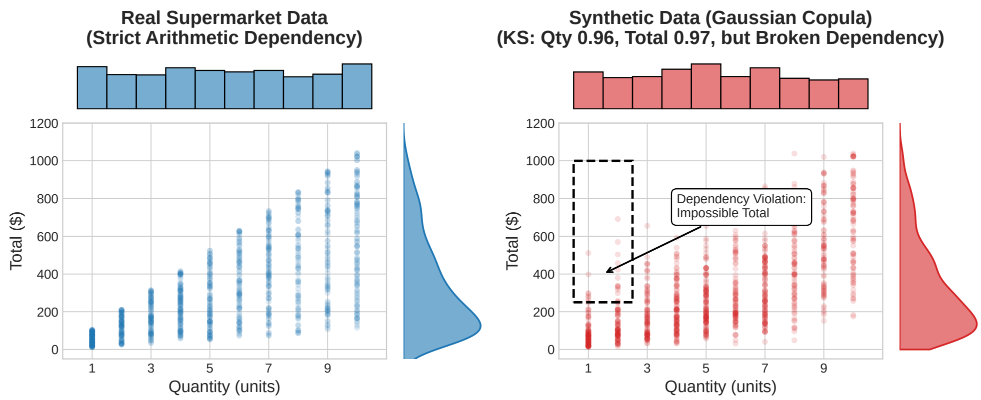

Fig. 1 An empirical demonstration of the Dependency Gap on the Supermarket Sales dataset. While generative models (e.g., Vine Copulas) successfully replicate the 1D marginal distributions of *Quantity* and *Total* (side histograms, KS ≈ 0.96), they fail to preserve the joint arithmetic constraint ($\text{Total} = \text{Price} \times \text{Quantity} \times (1 + \text{Tax})$). The synthetic cohort hallucinates impossible records, such as massive totals for tiny quantities (dashed box), which aggregate marginal metrics do not capture.

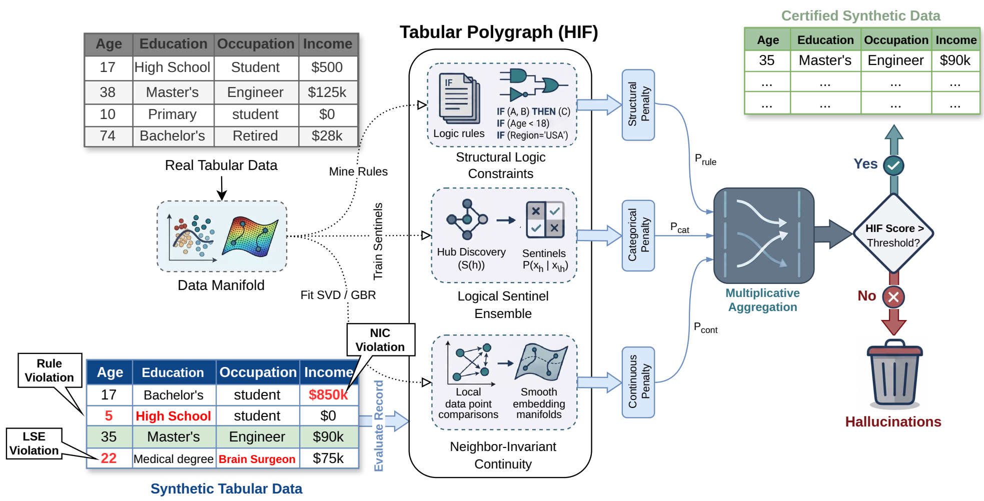

Fig. 2 Architecture of the Hybrid Integrity Framework (HIF). Synthetic tabular data is verified by the Logical Sentinel Ensemble (LSE) and Neighbor-Invariant Continuity (NIC) modules, which are combined via multiplicative aggregation to filter semantically inconsistent records.

[5] to neural architectures such as *CTGAN*, and shows that the Dependency Gap recurs across diverse generative paradigms in our tested settings.

Evaluation Metrics and the Dependency Gap.

Traditional evaluation pipelines primarily rely on statistical fidelity (e.g., Kolmogorov-Smirnov tests, Correlation Distance) and predictive utility (Train-on-Synthetic, Test-on-Real; TSTR). Although effective for global assessment, these metrics evaluate aggregate column-level statistics and are therefore blind to row-level conditional dependency violations—by design, a corruption that samples each feature from its correct marginal but ignores inter-feature correlations is entirely invisible to such metrics. Long et al. (2025) [6] were among the first to formalize these "Dependency Gaps," proposing rule-based audits for inter-column relationships. Umesh et al. [19] likewise studied the preservation of logical dependencies in synthetic data generation. Similarly, Yu et al. (2025) [9] introduced the SHAP Distance to evaluate fidelity through an explainability-aware lens. Notably, distribution-level auditing faces provable limits: identity and uniformity tests require sample sizes that scale with the domain support [20–22], a cost only partially mitigated even for product-structured distributions [23], and conditional independence testing is likewise sample-intensive in high dimensions [24]. HIF builds on these insights but automates dependency discovery through predictive synergy scoring. Unlike rule-mined audits, which are binary and require manual specification, HIF provides a soft probabilistic integrity score using non-compensatory multiplicative aggregation that prevents high-fidelity feature dimensions from masking localized dependency violations.

Probabilistic Dependency Auditing.

The automated discovery of inter-feature dependencies for data quality auditing is an active research area. Karabulut et al. (2025) [25] explored association rule mining to extract interpretable dependencies from large tables, while Yu et al. (2026) [26] demonstrated that dependency-aware representations enhance tabular understanding. HIF situates itself in this landscape as a fully probabilistic auditing framework: its LSE and NIC components are discriminative statistical models (random forests, gradient boosting regressors) trained on the real data distribution to score the conditional plausibility of synthetic records. Crucially, a fundamental theoretical distinction exists between HIF and general-purpose ensemble outlier detectors (e.g., Isolation Forests, Local Outlier Factors). Standard ensemble outlier detectors identify records situated in low-density feature space (extreme numerical or categorical outliers). In contrast, HIF is specifically designed to audit structural hallucinations ("impossible junctions") where every individual feature value is unexceptional and high-density, yet their joint combination violates physical or logical laws. Furthermore, while conventional anomaly ensembles aggregate member scores additively (averaging), HIF introduces a non-compensatory multiplicative product aggregation ($\prod$) that prevents well-behaved feature dimensions from compensating for localized dependency ruptures.

3 Mathematical Foundation

The Hybrid Integrity Framework (HIF) formalizes synthetic data auditing as a probabilistic conditional scoring task. Let $\mathcal{D}_{real}$ denote the original training dataset. We formally define the **ground-truth manifold** $\mathcal{M}_{real}$ as the support of the joint distribution $P_{real}(\mathbf{x})$, representing the localized subspace of valid feature configurations that occur in reality [27].

We define a synthetic record $\mathbf{x} = [\mathbf{x}_{cat}, \mathbf{x}_{cont}]$ (where $\mathbf{x}_{cat}$ and $\mathbf{x}_{cont}$ denote the subvectors of categorical and continuous features, respectively) as a *dependency-violating record* if it is plausible under the marginal distribution of individual features yet falls outside the bounds of the ground-truth manifold (i.e., it is highly improbable under the joint conditional distribution $P_{\mathcal{D}_{real}}(\mathbf{x}_h \mid \mathbf{x}_{\setminus h})$ learned from the data). This distinction is central: aggregate fidelity metrics (KS, TVD, JCD) assess whether synthetic records fall within the expected marginal range of each column in isolation; HIF additionally evaluates whether the joint configuration of each record structurally aligns with the ground-truth manifold. A record failing the latter criterion is a dependency violation even if every individual feature value is marginal-plausible. The complete calibration process for establishing these data-driven manifold boundaries is detailed in Algorithm 1.

3.1 Logical Sentinel Ensemble (LSE)

To audit categorical consistency, we first identify the set of Dependency Hubs $\mathcal{H} \subset \{1, \ldots, K_{hub}\}$ that govern the structural dependencies of the dataset. For a given row $\mathbf{x}$, let $x_h \in \mathcal{X}_h$ denote the categorical value of a target feature h, and let $\mathbf{x}_{\setminus h}$ denote the values of all other remaining features (i.e., $\mathbf{x}$ with feature h removed).

We train an independent Random Forest classifier [28]—leveraging decision tree recursive partitioning to capture non-linear feature interactions [29]—which we call a *Sentinel* and denote as f_h, to predict x_h given $\mathbf{x}_{\setminus h}$. The sentinel outputs a conditional probability distribution $P_{f_h}(x_h = c \mid \mathbf{x}_{\setminus h})$ for all possible categories $c \in \mathcal{X}_h$. We define the importance of a feature h using the Feature Dependency Score $S(h)$, which is the cross-validated classification accuracy of its sentinel:

$$S(h) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{real}} \left[\mathbb{I} \left(\arg\max_{c \in \mathcal{X}_h} P_{f_h}(x_h = c \mid \mathbf{x}_{\setminus h}) = x_h \right) \right] \tag{1}$$

where $\mathbb{I}$ is the indicator function. By maximizing $S(h)$, we select hubs that are most strongly constrained by conditional interactions with other attributes. While computationally efficient, this reliance on single-variable classification accuracy primarily captures univariate dependencies and may under-represent higher-order parity relationships. Future work will investigate Total Correlation and multi-way interaction kernels to broaden hub selection capability.

Running Example: Categorical Dependencies.

Consider a demographic dataset where $\mathbf{x}$ represents a person. Let x_h be their *Education Level* and $\mathbf{x}_{\setminus h}$ contain their *Age* (e.g., 14). Because a 14-year-old cannot hold a PhD in the real world, the sentinel f_h learns from the ground-truth data to assign $P_{f_h}(\text{PhD} \mid \text{Age} = 14) \approx 0$. If a generative model hallucinates a 14-year-old with a PhD, the sentinel will flag the record because the assigned probability for the hallucinated category is extremely low.

Confidence Floor Calibration.

Drawing inspiration from distribution-free quantile calibration in inductive conformal prediction [30–32], we establish a data-driven Confidence Floor δ_h for each hub based on empirical calibration error bounds. We define δ_h using the empirical 5th-percentile ($Q_{0.05}$) of the probability mass assigned to the correct category by the sentinel in the ground-truth data, with a safety minimum of 0.01:

$$\delta_h = \max\Big(Q_{0.05}\Big(\{P_{f_h}(x_h \mid \mathbf{x}_{\setminus h}) \mid \mathbf{x} \in \mathcal{D}_{real}\}\Big), 0.01\Big) \tag{2}$$

This 5th-percentile floor acts as a conservative safety margin. In our running example, if the real data occasionally contains 18-year-olds with college degrees, the sentinel's probability might be low but non-zero (e.g., 0.05). The floor δ_h ensures we do not penalize rare but valid records that fall within the natural density of the ground-truth manifold.

The per-hub penalty $\mathcal{P}_h(\mathbf{x})$ uses a soft-threshold activation to suppress noise-level deviations, harshly penalizing only those records whose probabilities fall far below the confidence floor (such as the impossible 14-year-old PhD):

$$\mathcal{P}_h(\mathbf{x}) = \text{clip}\left(\frac{\delta_h - P_{f_h}(x_h \mid \mathbf{x}_{\setminus h})}{\delta_h}, 0, 1\right) \tag{3}$$

The categorical penalty aggregates multiplicatively across hubs, ensuring that ruptures in multiple manifold dimensions are compounded.

$$\mathcal{P}_{cat}(\mathbf{x}) = 1 - \prod_{h \in \mathcal{H}} (1 - \mathcal{P}_h(\mathbf{x})) \tag{4}$$

For a rigorous theoretical analysis, including the proof of intersection soundness (Theorem 1), asymptotic statistical convergence (Theorem 2), and the upper bound on Type I false-rejection error under quantile calibration (Theorem 3), we refer the reader to Appendix A.

3.2 Neighbor-Invariant Continuity (NIC)

For continuous features $\mathbf{y}$, we evaluate integrity via Semantic Adjacency. We map the categorical context to a continuous latent space $\mathcal{Z}$ using a standardized spectral projection $\phi : \mathcal{X}_{cat} \to \mathcal{Z}$—drawing on classical Multiple Correspondence Analysis (MCA) and Singular Value Decomposition (SVD) on binary indicator matrices [33, 34]. Standardizing the indicator matrices by their standard deviation prior to SVD mathematically replicates the inverse-frequency weighting of MCA, ensuring that rare but highly informative categorical constraints dominate the continuous manifold projection. We then model the conditional expectation of each continuous feature y as a non-linear function of this manifold using a Gradient Boosting Regressor r_y [35, 36]:

$$\hat{y}(\mathbf{x}) = \mathbb{E}[y \mid \phi(\mathbf{x}_{cat})] \approx r_y(\phi(\mathbf{x}_{cat})) \tag{5}$$

Algorithm 1 HIF Manifold Calibration

Input: Ground-truth data $\mathcal{D}_{real}$, Hub count K_{hub}
Output: Sentinels $\{f_h\}$, Floors $\{\delta_h\}$, NIC Regressors $\{r_y\}$, Thresholds $\{\tau_y\}$

$\mathcal{H} \leftarrow$ Select top K_{hub} features by Feature Dependency Score $S(h)$
for each hub $h \in \mathcal{H}$ **do**
  Train Sentinel $f_h : \mathcal{X}_{\setminus h} \to [0,1]^{|\mathcal{X}_h|}$ on $\mathcal{D}_{real}$ to predict x_h
  $\delta_h \leftarrow \max(\text{Percentile}_5(\{f_h(\mathbf{x}_h \mid \mathbf{x}_{\setminus h}) \mid \mathbf{x} \in \mathcal{D}_{real}\}), 0.01)$
end for
$\mathcal{Z} \leftarrow \text{SVD}(\text{Scale}(\text{OneHot}(\mathcal{D}_{cat})))$ ▷ Non-Centered Spectral Manifold
for each continuous feature y **do**
  Train Gradient Boosting regressor $r_y : \mathcal{Z} \to \mathbb{R}$ on $\mathcal{D}_{real}$
  $\mathcal{E}_y \leftarrow \{|y - r_y(\mathbf{z})| \mid \mathbf{x} \in \mathcal{D}_{real}\}$ ▷ Ground-truth residuals
  $\tau_y \leftarrow \max(Q_{0.95}(\mathcal{E}_y), \text{Median}(\mathcal{E}_y) + 2 \cdot \text{MAD}(\mathcal{E}_y))$ ▷ Robust Threshold
  $\gamma_y \leftarrow \max(Q_{0.98}(\mathcal{E}_y), 2 \cdot \tau_y, 3 \cdot \text{MAD}(\mathcal{E}_y), 0.1)$ ▷ Dynamic Gamma Scaling
end for
$\mathcal{R} \leftarrow \text{MineImplicationRules}(\mathcal{D}_{real})$ ▷ Apriori-style, min_conf=0.95, min_supp=0.005

For any given record $\mathbf{x}$, let $e_y(\mathbf{x}) = |y(\mathbf{x}) - \hat{y}(\mathbf{x})|$ denote the absolute regression residual (i.e., the error between the actual continuous value and the expected value given the categorical context). We define the set of all ground-truth residuals as $\mathcal{E}_y = \{e_y(\mathbf{x}) \mid \mathbf{x} \in \mathcal{D}_{real}\}$.

Running Example: Continuous Dependencies.

Continuing our demographic example, suppose a synthetic 45-year-old Senior Executive is generated with an Income of \$12. The NIC regressor r_y looks at the categorical profile and predicts an expected income of $\hat{y} \approx \$120,000$. The absolute residual $e_y(\mathbf{x}) = \$119,988$ is astronomically high compared to the natural variance of executives in the real dataset, signaling a continuous dependency gap.

To handle extreme feature skewness (common in census domains), we define a robust hybrid threshold τ_y based on the ground-truth residuals $\mathcal{E}_y$. We use the Median Absolute Deviation (MAD), which provides a robust estimate of local manifold spread that is less sensitive to outliers than the standard deviation:

$$\tau_y = \max\left(Q_{0.95}(\mathcal{E}_y), \text{Median}(\mathcal{E}_y) + 2 \cdot \text{MAD}(\mathcal{E}_y)\right) \tag{6}$$

This threshold ensures that the NIC layer remains stable even in high-skew distributions, such as macroeconomic wage data.

The continuous penalty $\mathcal{P}_{cont}(\mathbf{x})$ is expressed using a dynamic scaling factor γ_y, formally defined as $\gamma_y = \max(Q_{0.98}(\mathcal{E}_y), 2\tau_y, 3\text{MAD}(\mathcal{E}_y), 0.1)$. This parameter defines the bounds of the penalty, scaling the response relative to the natural noise floor of the feature. This ensures that small deviations beyond τ_y result in soft penalties, while massive violations (like the \$12 executive) yield a maximum penalty of 1.0:

$$\mathcal{P}_{cont}(\mathbf{x}) = \max_y \left[\text{clip}\left(\frac{e_y(\mathbf{x}) - \tau_y}{\gamma_y}, 0, 1\right)\right] \tag{7}$$

As a defensive guard, NIC additionally detects *collapsed* continuous features whose conditional predictions have near-zero variance (i.e., the feature is effectively constant given the categorical context); for such degenerate features, all records receive a uniform base penalty, since the expected residual distribution provides no reliable baseline. This guard did not activate on any benchmark dataset in our experiments.

3.3 Multiplicative Aggregation

In contrast to additive fidelity metrics that allow a model to "compensate" for logical failures with high statistical overlap, HIF formalizes integrity as an Algebraic Intersection. Drawing from the structured drift-proof framework [10], the hybrid integrity score $H(\mathbf{x})$ is defined as the geometric mean of the component validities:

$$H(\mathbf{x}) = \left(\prod_{l \in \{cat, cont, rule\}} (1 - \mathcal{P}_l(\mathbf{x}))\right)^{1/L} \tag{8}$$

where l iterates over the active auditing layers. The geometric mean preserves the non-compensatory property of a Logical AND gate: a significant failure in any single manifold dimension ($\mathcal{P}_l \to 1$) drives the entire record integrity toward zero, regardless of how well the other components align (see empirical validation of this non-compensatory advantage in Table 1, Section 4.1).

Theoretical Scope: For a formal analysis of this non-compensatory aggregation logic under explicit assumptions, including intersection soundness (Theorem 1) and asymptotic convergence arguments for the empirical penalty functions (Theorem 2), we direct the reader to **Appendix A**.

While HIF prioritizes soft probabilistic validities via LSE and NIC, we incorporate symbolic implication rules as an optional *Structural Logic Constraint* layer ($l = rule$) to enforce known deterministic or physical laws (e.g., Age $\geq$ 18 for voters). This hybrid architecture allows the HIF to bridge the gap between differentiable manifolds and rigid symbolic constraints.

The Integrity Frontier.

The dataset-level HIF score is the expected record-level integrity. To provide a discrete diagnostic, a record is classified as a *Semantic Violation* if $H(\mathbf{x}) < 0.5$, defining the framework's Integrity Frontier.

$$\text{HIF Score} = \frac{1}{N}\sum_{i=1}^{N} H(\mathbf{x}_i) \tag{9}$$

The complete end-to-end auditing procedure for scoring a synthetic dataset is detailed in Algorithm 2.

Algorithm 2 HIF Synthetic Data Audit

Input: Synthetic record $\mathbf{x}$, Trained components from Alg 1
Output: Record Integrity Score $H(\mathbf{x})$

```
Step 1: Logical Sentinel Audit (Categorical)
for each hub h ∈ H do
    P_h ← clip((δ_h − f_h(x_h | x_\h)) / δ_h, 0, 1)
end for
P_cat ← 1 − ∏_{h∈H} (1 − P_h)          ▷ Multiplicative violation aggregation
Step 2: Neighbor-Invariant Continuity Audit (Continuous)
z ← φ(x_cat)                             ▷ Apply learned standardized projection
for each continuous feature y do
    ρ_y ← |y − r_y(z)|
    P_y ← clip((ρ_y − τ_y) / γ_y, 0, 1)
end for
P_cont ← max_y P_y
Step 3: Integrity Aggregation
P_rule ← 1 if x violates any r ∈ R else 0
H(x) ← (∏_l (1 − P_l))^{1/L} ∈ [0, 1]    ▷ Geometric mean of validities
```

In math notation:

- **Step 1: Logical Sentinel Audit (Categorical)**
- **for** each hub $h \in \mathcal{H}$ **do**
  - $\mathcal{P}_h \leftarrow \text{clip}\left(\frac{\delta_h - f_h(\mathbf{x}_h | \mathbf{x}_{\setminus h})}{\delta_h}, 0, 1\right)$
- **end for**
- $\mathcal{P}_{cat} \leftarrow 1 - \prod_{h \in \mathcal{H}}(1 - \mathcal{P}_h)$ ▷ Multiplicative violation aggregation
- **Step 2: Neighbor-Invariant Continuity Audit (Continuous)**
- $\mathbf{z} \leftarrow \phi(\mathbf{x}_{cat})$ ▷ Apply learned standardized projection
- **for** each continuous feature y **do**
  - $\rho_y \leftarrow |y - r_y(\mathbf{z})|$
  - $\mathcal{P}_y \leftarrow \text{clip}\left(\frac{\rho_y - \tau_y}{\gamma_y}, 0, 1\right)$
- **end for**
- $\mathcal{P}_{cont} \leftarrow \max_y \mathcal{P}_y$
- **Step 3: Integrity Aggregation**
- $\mathcal{P}_{rule} \leftarrow 1$ if $\mathbf{x}$ violates any $r \in \mathcal{R}$ else 0
- $H(\mathbf{x}) \leftarrow \left(\prod_l (1 - \mathcal{P}_l)\right)^{1/L} \in [0, 1]$ ▷ Geometric mean of validities

This score provides a bounded diagnostic $\in [0, 1]$, where 1.0 indicates a logically certified synthetic cohort.

3.4 Representation Stability and Fairness

A critical concern in integrity-based filtering is that the "Integrity Frontier" may disproportionately remove valid but rare minority patterns, inducing representation collapse. Because HIF penalizes records that violate *learned conditional dependencies* rather than marginal rarity—the confidence floor δ_h in Algorithm 1 is calibrated on the ground-truth manifold precisely to avoid penalizing rare-but-valid tail records—filtering operates on structural consistency rather than demographic support. A systematic quantitative audit of representation drift across sensitive attributes (e.g., Race, Sex) under integrity filtering is left to future work.

3.5 Theoretical Justification: Non-Compensatory Evaluation

A primary critique of multiplicative evaluation is that absolute scores often appear lower than the near-unity scores (≥ 0.95) typical of aggregate 1D distributional metrics in standard evaluation benchmarks [2, 6, 12]. This is a deliberate design property. In datasets governed by multiple simultaneous conditional constraints, a record with a severe violation of any single dependency should not receive a high integrity score, regardless of how well its remaining features conform to marginal expectations.

In standard aggregate metrics, a high-fidelity feature dimension can arithmetically compensate for localized dependency failures, inflating the global score. HIF avoids this through *non-compensatory aggregation*. Theoretically, assuming conditional independence of error dimensions given the feature representations, the product of validities $\prod(1-\mathcal{P}_l)$ serves as a pseudo-likelihood proxy for the joint structural conformity of the record. Furthermore, under empirical quantile calibration ($\delta_h = Q_\alpha$ and $\gamma_y = Q_{1-\beta}$), the false-rejection probability for valid tail records is upper-bounded by the Union Bound under stated assumptions (Theorem 3, Appendix A.3).

The geometric mean enforces a robust non-compensatory behavior: a significant penalty on any single auditing layer ($\mathcal{P}_l \to 1$) suppresses the overall record integrity toward zero. Concretely, if a generator produces a record whose continuous feature value is highly improbable under the conditional distribution for its categorical context (e.g., a wage value inconsistent with the learned conditional for that employment category), the NIC penalty dominates the geometric mean regardless of how well other features are aligned. This positions the HIF score as a conservative proxy for joint distributional consistency—not an average of per-column fidelities that can obscure isolated but severe dependency violations.

3.6 End-to-End Pipeline Walkthrough

To unify the notation across all components, consider a synthetic record $\mathbf{x} = [\mathbf{x}_{cat}, \mathbf{x}_{cont}]$ evaluated by HIF:

1. **Categorical Audit (LSE):** For a record with Age = 14 and Education = PhD, the sentinel f_h for hub h = Education evaluates $P_{f_h}(\text{PhD} \mid \text{Age} = 14) = 0.0001$. Given the calibrated floor $\delta_h = 0.05$, the categorical penalty evaluates to $\mathcal{P}_h = \text{clip}\left(\frac{0.05-0.0001}{0.05}, 0, 1\right) = 0.998$.
2. **Continuous Audit (NIC):** SVD projects $[\text{Age} = 14, \text{Education} = \text{PhD}]$ to latent vector $\mathbf{z}$. The regressor r_y predicts expected salary $\hat{y} = \$12,000$ with robust threshold $\tau_y = \$5,000$ and tolerance bandwidth $\gamma_y = \$10,000$. For a hallucinated salary $y = \$150,000$, the continuous penalty evaluates to $\mathcal{P}_y = \text{clip}\left(\frac{|150{,}000-12{,}000|-5{,}000}{10{,}000}, 0, 1\right) = 1.0$.
3. **Multiplicative Aggregation:** The overall record score collapses:

$$H(\mathbf{x}) = [(1-0.998) \times (1-1.0)]^{1/2} = (0.002 \times 0.0)^{1/2} = 0.000 \tag{10}$$

The record is flagged as a structural hallucination ($H < 0.5$).

4 Results

We evaluated the Hybrid Integrity Framework (HIF) on benchmark datasets representing diverse structural complexity: the U.S. Census ACS dataset [37, 38], Adult Income, Credit Card Default, Online Purchases, and Supermarket Sales datasets. Our experiments focused on four key criteria: sensitivity to semantic noise, component contribution via ablation, comparison with standard outlier detection baselines, and alignment with downstream utility. Experiments used $N = 3$–10 random seeds depending on the evaluation; we report mean $\pm$ standard deviation (SD) unless otherwise noted.

To calibrate the framework's response, we executed a controlled semantic violation protocol. Unlike random corruption, which breaks global distributions, we stochastically permuted (shuffled) the values of specific features across different rows within highly constrained categorical-continuous dependencies for a controlled fraction of records $\eta \in [0, 0.6]$ (Census ACS, $N = 3$ seeds). By shuffling existing values across rows rather than injecting external noise, this process preserves each column's marginal distribution (satisfying standard MM and KS metrics) while breaking the underlying joint conditional structure at the row level.

As shown in our multi-seed validation, the HIF score exhibited strong monotonicity in our runs (Spearman $\rho = -1.0$ across seeds) with respect to η: on Census ACS, the mean HIF decreased from 0.941 ± 0.003 at $\eta = 0$ to 0.712 ± 0.002 at $\eta = 0.6$. The marginal metrics—Moment Matching (MM) and Kolmogorov-Smirnov (KS)—remained effectively static ($\rho \approx 0$), since the shuffled values are drawn from the intact marginals. Notably, the Joint Correlation Distance (JCD) also decayed sharply under this protocol ($88.8 \rightarrow 65.0$, $\rho = -1.0$): shuffling a feature destroys its pairwise correlations, so JCD does respond to broad permutation-based corruption. JCD is therefore a sensitive aggregate drift detector, but it yields only a dataset-level summary and cannot attribute the degradation to specific records. This motivates HIF's row-level output and the low-rate, marginals-preserving conditional-swap protocol used for the differential diagnostic in Section 4.7.

Addressing Circularity in Evaluation.

We emphasize that this controlled corruption protocol is used strictly for metric response calibration (verifying monotonic decay under controlled error rates), rather than as the primary benchmark for diagnostic utility. To ensure non-circular evaluation, our main empirical validation assesses HIF across two independent regimes: (1) an evaluation on *held-out error types* (four generic corruption families that HIF was not specifically engineered to detect, plus the targeted dependency-violation family as a positive control; Section 4.2), and (2) downstream predictive utility auditing on unmanipulated synthetic cohorts produced by real generative architectures (CTGAN, Vine Copulas) (Section 4.4).

Evaluation Protocol and Threats to Validity.

We distinguish ranking quality from thresholded classification quality. Accordingly, ROC-AUC and PR-AUC are treated as threshold-independent diagnostics of error ranking, while F1 is reported as a threshold-dependent operating-point metric. For baseline comparability, IF and LOF use a contamination parameter set to the known corruption rate in synthetic stress tests; this favors those baselines for F1 and is reported explicitly to avoid hidden calibration advantages. We also separate calibration experiments from deployment-style experiments: targeted corruption tests are used to probe metric response properties, whereas held-out error families and unmanipulated generator outputs are used to assess external validity. Residual threats include dataset selection bias and the possibility that rare-but-valid tail patterns are partially coupled with high-penalty regions.

4.1 Component Ablation Study

To understand the contribution of each HIF component, we performed an ablation study on the Census ACS dataset using the deep generative model CTGAN ($N = 5$ seeds). As shown in Table 1, the ablation reveals the distinct roles of each filtering module on downstream utility. Because *Census ACS* is dominated by complex, non-linear categorical dependencies, the Logical Sentinel Ensemble (LSE) is the most critical individual component, raising F1 from 0.395 ± 0.022 (no filtering) to 0.782 ± 0.071 while retaining 26.8% of records. In contrast, simple rule-based filtering (Rules-only) removes few records (2.8% violation rate) and leaves utility near baseline, and Neighbor-Invariant Continuity alone (NIC-only) barely prunes anything (12.4% violation rate), since Census ACS numeric features carry few categorical-driven conditional failures. Combining LSE with NIC yields intermediate behavior (retention 41.6%, F1 0.674 ± 0.091).

Crucially, the ablation illustrates the aggregation method's role: Full HIF with multiplicative product aggregation ($F1 = 0.546 \pm 0.072$) outperforms the arithmetic mean ($F1 = 0.405 \pm 0.037$) on the CTGAN cohort. The multiplicative product acts as a non-compensatory (Logical AND) gate—preventing a high score in one dimension from hiding a severe failure in another—pruning an additional 30% of records relative to arithmetic aggregation (retaining 64.0% vs. 94.8%). The magnitude of this advantage should be read with caution: retention and utility are confounded, and the gap is driven primarily by how aggressively each variant prunes. Consistent with this interpretation, the aggregation choice is immaterial in settings free of structural errors: under Gaussian Copula on Census ACS (violation rate $< 1\%$), all ablation configurations yield statistically indistinguishable F1 (0.934–0.941).

Table 1 Component Ablation Study on Census ACS (CTGAN, $N = 5$ seeds). The multiplicative product implements a non-compensatory Logical AND gate, pruning more records than the arithmetic mean and reaching the highest F1 on a cohort with structural errors. In cohorts free of structural errors (Gaussian Copula), all configurations are statistically indistinguishable (F1 0.934–0.941).

Ablation	Retention (%)	F1 (mean ± SEM)	Violation Rate
No filtering	100.0%	0.395 ± 0.022	0.0%
LSE-only	26.8%	0.782 ± 0.071	73.2%
NIC-only	87.6%	0.373 ± 0.015	12.4%
Rules-only	97.2%	0.396 ± 0.029	2.8%
LSE + NIC	41.6%	0.674 ± 0.091	58.4%
Full HIF (arith.)	94.8%	0.405 ± 0.037	5.2%
Full HIF (multiplicative)	**64.0%**	**0.546 ± 0.072**	**36.0%**

4.2 Comparison with Outlier Detection Baselines

A natural question is whether standard outlier detection methods—specifically Isolation Forest (IF) and Local Outlier Factor (LOF)—can serve as alternatives to HIF for identifying synthetic data flaws. Table 2 compares detection performance across five error types on Census ACS at 40% corruption level ($N = 3$ seeds).

We report both threshold-independent ROC-AUC and standard F1 scores. The distinction between metrics is crucial: LOF and IF are configured with a `contamination` parameter set to the known corruption rate (0.4), giving them a tuned decision threshold for F1 in this stress-test setup, whereas HIF uses a fixed non-parametric default threshold ($H > 0.5$).

Under threshold-independent ROC-AUC, HIF strongly outperforms baseline detectors on *Conditional Dependency Violations* (ROC-AUC = 0.837, PR-AUC = 0.799), whereas Isolation Forest is near-random on marginal-preserving dependency swaps (ROC-AUC = 0.519) and LOF achieves lower ranking quality (ROC-AUC = 0.728, PR-AUC = 0.663). Conversely, generic outlier detectors excel at catching broad geometric feature outliers (e.g., random noise injection), where LOF reaches ROC-AUC = 0.864. HIF also maintains high sensitivity on random noise (ROC-AUC = 0.900) because breaking marginals simultaneously disrupts conditional dependencies. On distribution-level shifts (covariate shift), Isolation Forest outperforms HIF (ROC-AUC 0.488 vs. 0.283), indicating that HIF is a specialized diagnostic for row-level joint conditional integrity rather than global distribution drift.

Why do standard outlier detectors fail on Dependency Gaps (Semantic Hallucinations)? Methods like Isolation Forest and LOF evaluate geometric distance or marginal density across the entire continuous feature space. If a synthetic record hallucinates an impossible logic combination (e.g., a 14-year-old with a PhD), but '14' and 'PhD' are both common values marginally, the record resides in a dense region of the globally projected feature space. Because Isolation Forest splits features independently, it rarely isolates the specific interacting dimensions that violate the conditional rule. A Dependency Gap is a structural, semantic error, not a geometric density anomaly, which is precisely why HIF's logic-aware sentinels succeed where geometric detectors fail.

Table 2 Held-Out Error Detection: HIF vs. Outlier Baselines (Census ACS, 40% corruption). We report threshold-independent ROC-AUC and F1 score across 3 random seeds. HIF achieves strong discrimination on conditional dependency violations (ROC-AUC = 0.837) where Isolation Forest is near-random (ROC-AUC = 0.519). Geometric detectors (LOF, IF) excel at broad marginal noise where HIF is intentionally permissive.

	ROC-AUC (↑)			**F1 Score (↑)**		
Error Type	**HIF**	**IF**	**LOF**	**HIF**	**IF**	**LOF**
Random Noise Injection	**0.900**	0.520	0.864	0.385	0.430	**0.616**
Row Duplication	0.508	0.495	**0.518**	0.019	0.407	**0.541**
Feature Dropout	**0.659**	0.275	0.489	0.070	0.157	**0.518**
Covariate Shift	0.283	**0.488**	0.212	0.008	**0.401**	0.286
Conditional Dependency Violations	**0.837**	0.519	0.728	0.225	0.415	**0.597**

4.3 Statistical Significance of Utility Preservation

Table 3 reports downstream utility preservation on Census ACS (binary median-split `household_income` target, $N = 10$ seeds). For the Gaussian Copula, a paired t-test between full synthetic and HIF-filtered records yields $p = 0.133$ (95% CI for F1 change: $[-0.008, +0.001]$), indicating no statistically detectable utility degradation at $\alpha = 0.05$ for this setting. This result is expected: copula cohorts on Census ACS are largely free of conditional dependency violations (violation rate $< 1\%$), so HIF's specialized filtering has little to act on. For deep generative models (CTGAN), downstream utility gains are statistically significant (paired t-test $p = 0.002$; Wilcoxon signed-rank $p = 0.002$; 95% CI for F1 improvement: $[+0.062, +0.188]$), with absolute F1 improving from 0.432 ± 0.024 to 0.557 ± 0.044 (mean improvement $+0.125$). This supports the claim that HIF is particularly useful for architectures that produce joint dependency violations.

Table 3 Statistical significance on Census ACS ($N = 10$ seeds, binary median-split `household_income` target). Paired t-test for Gaussian Copula: $p = 0.133$ (no detectable utility change at $\alpha = 0.05$). For CTGAN, utility improvement is statistically significant ($p = 0.002$, 95% CI: $[+0.062, +0.188]$).

Generator	Variant	Retention (%)	F1 (mean ± SEM)
Gaussian Copula	Full synthetic	100.0%	0.940 ± 0.002
Gaussian Copula	Rule-only	99.1%	0.937 ± 0.002
Gaussian Copula	**HIF Filtered**	**99.2%**	**0.937 ± 0.002**
CTGAN	Full synthetic	100.0%	0.432 ± 0.024
CTGAN	Rule-only	95.9%	0.418 ± 0.025
CTGAN	**HIF Filtered**	**64.2%**	**0.557 ± 0.044**

4.4 Alignment with Downstream Utility

A critical requirement for any synthetic data metric is its ability to act as a proxy for downstream predictive utility, a validation paradigm now standard in privacy-sensitive domains [39]. We measured downstream TSTR F1 under integrity filtering across representative generative architectures (Table 4), using a Random Forest classifier [28] implemented via *Scikit-learn* [40] for utility tasks. The *Rule-only Baseline* prunes records that violate any mined implication rules ($r \in \mathcal{R}$), whereas the *HIF Filtered* variant uses the hybrid semantic frontier ($H < 0.5$).

Table 4 demonstrates the practical impact: filtering a synthetic cohort to retain high-integrity records maintains or improves downstream predictive power, with the direction and magnitude governed by whether structural errors are present. On Census ACS with CTGAN (violation rate 32%), HIF filtering yields a statistically significant +12.5 points F1 gain (0.557 vs. 0.432, Section 4.4). On cohorts largely free of structural errors—Gaussian Copula on Census ACS (violation rate $< 1\%$)—filtering preserves utility (change within ± 0.003). On Online Purchases with the Gaussian Copula (mild structural errors, 42% violation rate), filtering retains 57.8% of records and F1 moves within noise ($0.983 \rightarrow 0.973$, $N = 3$ seeds), tempering the hypothesis that dependency violations always act as predictive noise: whether removing them improves utility depends on the strength of the violation-to-label association.

Table 4 Downstream utility filtering. Selecting records that satisfy the manifold laws maintains or improves predictive performance while pruning logically inconsistent records. Gaussian Copula rows on Census ACS and CTGAN rows on Census ACS use $N = 10$ seeds; Online Purchases Gaussian Copula row uses the cross-architecture benchmark ($N = 3$ seeds).

Dataset	Generator	Variant	Retention (%)	F1 (± SD)	Accuracy (± SD)
ACS (Census)	Gaussian Copula	Full synthetic	100.0	0.940 ± 0.006	0.940 ± 0.006
	Gaussian Copula	**HIF Filtered**	**99.2**	0.937 ± 0.006	0.937 ± 0.006
	CTGAN	Full synthetic	100.0	0.432 ± 0.075	0.530 ± 0.039
	CTGAN	**HIF Filtered**	**64.2**	**0.557** ± 0.139	**0.620** ± 0.087
Purchases	Gaussian Copula	Full synthetic	100.0	0.983 ± 0.003	–
	Gaussian Copula	**HIF Filtered**	**57.8**	0.973 ± 0.013	–

4.5 Cross-Architecture Maturity Audit and Cohort Certification

To evaluate the diagnostic utility of HIF across generative paradigms, we conducted a cross-architecture audit across four benchmark datasets from diverse domains (Table 5): Supermarket Sales (retail arithmetic constraints), Online Purchases (e-commerce transaction constraints), Credit Card Default (financial classification), and Adult Census Income (demographic benchmark).

The results reveal a key insight into synthetic data evaluation:

1. Integrity-Fidelity Decoupling on Constrained Domains.

Across all four datasets, Vine Copulas achieve high marginal statistical fit (KS $\geq$ 0.967). However, on datasets with cross-feature relational constraints (Supermarket Sales and Online Purchases), Vine Copulas exhibit severe joint dependency failures, with violation rates of **85.4% to 99.8%** (HIF $\leq$ 0.278). CTGAN and TVAE exhibit similar dependency deficits on transaction constraints (HIF = 0.02–0.29, violation rates 85.4%–100%). Marginal metrics (KS) report these models as high-fidelity, underscoring that aggregate fidelity metrics do not capture joint dependency ruptures.

2. Evaluation on Unconstrained Manifolds.

On unconstrained demographic and financial benchmarks (Adult Income, Credit Default), raw generators preserve conditional structures better, with HIF scores ranging from 0.30 to 0.98 (violation rates 0.8%–72.4%). In these regimes, HIF provides a continuous gradient of structural integrity, allowing practitioners to compare generator maturity. Notably, TVAE achieves the highest HIF on both Credit Default (0.835) and Adult Income (0.981), demonstrating that VAE-based architectures can excel at preserving joint dependencies on unconstrained domains.

3. Quantitative Decoupling Evidence.

Pearson correlation analysis across all 16 dataset–generator configurations characterizes the decoupling precisely: HIF shows no linear relationship with marginal fidelity ($\rho(\text{HIF}, \text{KS}) = 0.008$) and only a weak relationship with support coverage ($\rho(\text{HIF}, \alpha) = 0.275$). In contrast, HIF exhibits a strong positive correlation with joint correlation fidelity ($\rho(\text{HIF}, \text{JCD}) = +0.83$). This indicates that HIF aligns with joint-dependency structure while being decoupled from per-column marginal distributional metrics: high marginal fidelity (KS) does not imply high integrity, exactly the pattern observed for Vine Copulas on constrained domains.

Table 5 Cross-Architecture Audit: Statistical Fidelity (KS), Support Overlap (α-Precision, β-Recall), and Joint Dependency Integrity (HIF) across benchmark datasets ($N = 3$ seeds, Mean ± SD). High marginal fidelity (KS > 0.96) or high support coverage (α-Prec > 0.79) does not imply joint conditional integrity. On datasets with cross-feature constraints, models exhibit up to 100% dependency violation rates.

Generator	KS (↑)	α-Prec (↑)	β-Rec (↑)	HIF (↑)	Viol. Rate (↓)
Supermarket Sales					
Gaussian Copula	0.902 ± 0.005	0.895 ± 0.008	0.923 ± 0.009	**0.968 ± 0.003**	**2.1 ± 0.4%**
Vine Copula	**0.973 ± 0.003**	0.790 ± 0.018	0.845 ± 0.009	0.025 ± 0.003	99.8 ± 0.2%
CTGAN	0.821 ± 0.015	0.705 ± 0.059	0.871 ± 0.017	0.030 ± 0.002	100.0 ± 0.0%
TVAE	0.887 ± 0.026	0.852 ± 0.060	0.962 ± 0.002	0.022 ± 0.002	100.0 ± 0.0%
Online Purchases					
Gaussian Copula	0.753 ± 0.012	0.491 ± 0.003	0.829 ± 0.021	**0.626 ± 0.002**	**42.2 ± 2.0%**
Vine Copula	**0.967 ± 0.003**	0.840 ± 0.016	0.943 ± 0.009	0.278 ± 0.003	85.4 ± 0.7%
CTGAN	0.660 ± 0.089	0.523 ± 0.069	0.909 ± 0.061	0.285 ± 0.014	85.4 ± 4.9%
TVAE	0.860 ± 0.011	0.722 ± 0.021	0.813 ± 0.020	0.125 ± 0.008	99.3 ± 0.8%
Credit Default					
Gaussian Copula	0.745 ± 0.000	0.721 ± 0.001	0.956 ± 0.009	0.611 ± 0.006	30.8 ± 0.0%
Vine Copula	**0.981 ± 0.002**	0.800 ± 0.015	0.819 ± 0.012	0.390 ± 0.004	57.7 ± 0.2%
CTGAN	0.705 ± 0.024	0.571 ± 0.093	0.877 ± 0.016	0.296 ± 0.049	72.4 ± 6.3%
TVAE	0.886 ± 0.007	0.965 ± 0.011	0.918 ± 0.010	**0.835 ± 0.003**	**13.1 ± 0.5%**
Adult Income					
Gaussian Copula	0.847 ± 0.005	0.936 ± 0.006	0.838 ± 0.009	0.824 ± 0.001	11.6 ± 0.8%
Vine Copula	**0.980 ± 0.007**	**0.962 ± 0.006**	0.799 ± 0.029	0.761 ± 0.007	17.1 ± 1.4%
CTGAN	0.695 ± 0.029	0.796 ± 0.044	0.799 ± 0.043	0.657 ± 0.010	30.9 ± 2.0%
TVAE	0.893 ± 0.012	0.655 ± 0.029	0.729 ± 0.016	**0.981 ± 0.005**	**0.8 ± 0.6%**

4.6 Qualitative Analysis: Relational Subtotal Violations

A qualitative audit of the dependency violations identified by HIF on the Supermarket Sales dataset (Vine Copula) confirmed systemic relational inconsistencies. In the synthetic cohort, generative models match the marginal distribution of individual columns (`unit_price`, `quantity`, `subtotal`) in isolation but fail to respect the arithmetic dependency $\text{subtotal} = \text{unit\_price} \times \text{quantity} \times (1 + \text{tax})$, even though each column value falls within its valid numeric range. HIF flagged essentially the entire cohort (violation rate 99.8% across $N = 3$ seeds), with the Logical Sentinel Ensemble and the mined implication rules contributing the bulk of the penalty mass (component violation rates 99.9% and 63.8%, respectively), while the NIC continuous layer remained near-inactive on this domain (0.0%). While standard KS tests and joint correlation metrics ignore these violations because each column value falls within its valid marginal range, HIF's row-level penalty localizes the records that carry the arithmetic ruptures, demonstrating the necessity of row-level conditional auditing.

4.7 Differential Diagnostic: Integrity vs. Loyalty

To determine whether HIF provides unique utility over joint statistical metrics, we evaluated its response at low corruption levels ($p \in [0.001, 0.1]$) using a conditional-swap protocol that preserves marginal distributions (Census ACS, $N = 3$ seeds). At these low rates, the HIF score decays monotonically with p (Spearman $\rho = -1.0$; from 0.941 at $p = 0$ to 0.918 at $p = 0.1$), whereas the aggregate marginal metrics (Moment Matching, KS) remain static and the Joint Correlation Distance moves only marginally ($88.8 \rightarrow 87.9$, relative shift $< 1.2\%$). The small absolute JCD change reflects dilution: a few thousand corrupted rows contribute negligibly to a cohort-averaged correlation statistic. Because HIF emits a per-record penalty, it localizes exactly which rows carry the corrupted conditional relationships—a capability no aggregate metric provides.

We emphasize a complementary reading of these results: the JCD is not insensitive to joint corruption in general. Under the broad, high-rate permutation protocol described earlier in this section (which destroys pairwise correlations), the JCD decays sharply in lockstep with the HIF score. The two tools therefore address related but distinct failure modes: the JCD is a sensitive dataset-level drift detector, while HIF provides record-level attribution of conditional dependency violations. They are complements, not substitutes.

5 Discussion

The results indicate that the Hybrid Integrity Framework provides a conservative diagnosis in our experimental settings. By employing a multiplicative aggregation model, HIF ensures that a failure in any single semantic dimension results in a low integrity score. This non-compensatory evaluation is important for high-stakes domains, where a single inconsistent record can invalidate an analysis.

5.1 Reconciling Generative Paradigms: Statistical vs. Neural Trade-offs

A central finding of our cross-architecture audit (Table 5) is that the Dependency Gap is not restricted to deep neural black-boxes; rather, it appears as a recurring pathology affecting both statistical and neural generative models in distinct ways.

Furthermore, structural discrepancies across datasets stem directly from their underlying data-generating characteristics. Microdata benchmarks such as Census ACS are dominated by high-cardinality discrete categorical hubs (`household_income`, `housing_cost`, `poverty_status`, `education`, `tenure`) governed by strong conditional logic. In contrast, transactional domains (Supermarket Sales, Online Purchases) are governed by exact relational arithmetic ($\text{subtotal} = \text{unit\_price} \times \text{quantity} \times (1 + \text{tax})$).

In our runs, statistical copulas preserve integrity well on Census ACS and Supermarket Sales (HIF 0.94–0.97 for the Gaussian Copula), but degrade on Online Purchases and Credit Default (HIF 0.61–0.63), and Vine Copulas—despite the highest marginal fit (KS $\geq$ 0.967)—catastrophically violate transaction arithmetic constraints (HIF 0.02–0.28). Conversely, neural models such as CTGAN capture discrete categorical clusters on demographic data (HIF 0.30–0.66) but struggle with continuous arithmetic identities (HIF 0.03 on Supermarket Sales).

This reconciles our core motivation: the Dependency Gap does not stem from neural opacity alone, nor from copula rigidity. It is an inherent artifact of standard generative loss functions, which optimize aggregate distributional fit (likelihood, Wasserstein distance, or marginal discrimination) rather than row-level joint conditional integrity. Consequently, post-hoc row-level auditing is necessary regardless of whether the underlying generator is statistical or neural.

5.2 Practical Interpretation and Computational Scalability

Finally, we address the interpretation of conservative integrity scores. Because HIF computes the geometric mean of the component validities across L auditing layers, the overall score $H(\mathbf{x}) = \left(\prod_l \mathcal{C}_l\right)^{1/L}$ requires simultaneous joint satisfaction of all conditional models. This conservative behavior is an intentional design choice for high-stakes auditing where severe failure on any single conditional model should penalize the composite score.

The LSE architecture is highly scalable, with a computational complexity of $O(N \cdot K)$ for auditing N records across K hubs. The parallelizable nature of this design allows it to be scaled effectively to tables with thousands of columns. Our empirical sweep (Appendix C.1) shows stable performance even at $K = 1$, meaning the hub count can be minimized to reduce the K-multiplier overhead. In our benchmarks, auditing 10,000 synthetic rows required $\approx 1.5\,\mathrm{s}$ on standard multi-core commodity hardware. However, the memory overhead associated with training and storing an ensemble of K Random Forest sentinels for ultra-high-dimensional tables (e.g., $D > 10,000$) remains a potential hardware bottleneck. In such regimes, we propose replacing complex forest-based sentinels with sparse-linear or neuro-inspired models that offer a better trade-off between semantic depth and memory usage.

5.3 Specificity-Sensitivity Trade-off

A critical observation during our validation was the relative performance of HIF compared to global statistical metrics (e.g., JCD) under random corruption. While JCD exhibits higher sensitivity to broad marginal distributional drift, HIF prioritizes *dependency specificity*: it is calibrated to flag violations of learned conditional relationships, not marginal distributional jitter. This is a deliberate design choice. A generic anomaly detector sensitive to marginal outliers (such as LOF) would produce false positives on valid but rare records; HIF's sensitivity is bounded by the confidence floors learned from the real data distribution. This specificity–sensitivity trade-off is essential for high-stakes applications, where a dataset's utility depends on the preservation of inter-feature dependencies rather than the suppression of all marginal variation.

5.4 Limitations and Failure Regimes

Despite strong results, several limitations remain. First, HIF is not designed as a global covariate-shift detector and can underperform methods such as Isolation Forest on broad geometric drift. Second, the quality of hub selection and confidence-floor calibration depends on sample size and class balance; in sparse regimes, rare-but-valid records may overlap with high-penalty regions. Third, the fixed frontier ($H < 0.5$) is practical but not universally optimal; deployment settings may require task-specific threshold calibration and cost-aware tuning, and extending calibration to non-exchangeable data streams via conformal methods [41] remains open. Fourth, the current empirical benchmark set, while diverse, does not exhaust domain types (for example, temporal or relational tables), so external validity should be interpreted cautiously.

5.5 Broader Impact

By providing a verifiable integrity signal, HIF strengthens the case for reproducible research in privacy-sensitive domains. We recommend using HIF primarily as a diagnostic tool to guide generative model refinement rather than as a purely exclusionary filter.

Quantitative Privacy Audit.

To address the privacy implications of integrity filtering, we conducted a Membership Inference Attack (MIA) audit using distance-to-closest-record (DCR) shadow modeling (Appendix C.3). The AUC values remained near random-guessing behavior for both the full synthetic cohort ($AUC = 0.490 \pm 0.015$) and the HIF-filtered cohort ($AUC = 0.499 \pm 0.024$), indicating no clear membership leakage signal introduced by integrity filtering in this setup, even when measured against the stringent standards of modern differential privacy [42].

6 Conclusion

The generation of synthetic tabular data is pivotal for democratizing research in highly regulated sectors. However, as generative models grow in complexity, the metrics used to evaluate them must evolve beyond aggregate marginal alignment to assess row-level joint distributional consistency. In this study, we introduced the Hybrid Integrity Framework (HIF), a probabilistic diagnostic for detecting inter-feature dependency violations in synthetic tabular data. By combining the Logical Sentinel Ensemble (LSE), which learns conditional feature dependencies via discriminative classifiers, with Neighbor-Invariant Continuity (NIC), which scores continuous feature plausibility within the learned categorical context, HIF provides a technically rigorous foundation for identifying the "Dependency Frontier" of a synthetic cohort.

Our empirical results on Census ACS and transaction benchmarks show monotonic HIF degradation with increasing targeted dependency corruption in our experiments, while marginal metrics such as Moment Matching and KS remain static under the same corruption, indicating that they evaluate different failure modes; joint correlation metrics also respond to broad corruption, but only HIF attributes it to specific records. Our cross-architecture audit reveals a persistent "Integrity-Fidelity Decoupling": sophisticated models such as Vine Copulas can achieve high distributional fidelity (KS > 0.96) while failing 85–100% of dependency integrity tests on constrained datasets. By implementing non-compensatory probabilistic aggregation across independently learned dependency models, HIF provides researchers with a conservative row-level signal of joint distributional consistency—specifically, of conformity to the conditional feature dependencies of the real data—that aggregate marginal metrics often miss. We further show that HIF outperforms standard geometric outlier detectors (Isolation Forest, LOF) on conditional dependency violations while remaining appropriately insensitive to marginal noise types that do not constitute dependency failures. Future work will extend this framework to high-frequency financial time-series data [43] and explore kernel-based projections to capture higher-order dependencies [44–46]. Ultimately, HIF complements aggregate fidelity metrics by providing a specialized row-level dependency diagnostic, enabling more reliable synthetic datasets for high-stakes scientific research.

Code and Data Availability

The complete open-source implementation, experimental benchmarks, and dataset configurations are available anonymously at https://anonymous.4open.science/r/tabular-polygraph-21CE.

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A Formal Results for the HIF Score Under Stated Assumptions

We provide a formal analysis of Hybrid Integrity Framework (HIF) score properties under the **structured drift-proof (SDP)** framework introduced by Shang et al. [10]. The results below should be interpreted as assumption-scoped statements rather than unconditional guarantees.

A.1 Proof of Intersection Soundness

Theorem 1. The HIF score $H(\mathbf{x})$ defines a membership certificate for the intersection of semantic manifolds $\mathcal{M}_{total} = \bigcap_l \mathcal{M}_l$. Under the idealization that each component validity satisfies $\mathcal{C}_l(\mathbf{x}) = 1$ iff $\mathbf{x} \in \mathcal{M}_l$ and $\mathcal{C}_l(\mathbf{x}) \in [0, 1)$ otherwise, $H(\mathbf{x})$ attains its maximum value 1 exactly on $\mathcal{M}_{total}$:

$$\nVdash[H(\mathbf{x}) = 1] = \prod_l \nVdash[\mathbf{x} \in \mathcal{M}_l] \tag{11}$$

where L is the number of auditing layers and $H(\mathbf{x}) = \left(\prod_l \mathcal{C}_l(\mathbf{x})\right)^{1/L}$ is the geometric mean of the component validities, matching the implementation in Algorithm 2.

1. **Completeness:** If $\mathbf{x} \in \mathcal{M}_{total}$, then $\mathbf{x} \in \mathcal{M}_l$ for all l. Thus, $\mathcal{C}_l(\mathbf{x}) = 1$ for all l, and $H(\mathbf{x}) = \left(\prod_l 1\right)^{1/L} = 1$.
2. **Soundness (Non-Compensatory Property):** Suppose $\mathbf{x} \notin \mathcal{M}_{total}$. Then, there exists at least one l^* such that $\mathbf{x} \notin \mathcal{M}_{l^*}$, implying $\mathcal{C}_{l^*}(\mathbf{x}) < 1$. Because $0 \le \mathcal{C}_l(\mathbf{x}) \le 1$ for all l, the geometric mean satisfies $H(\mathbf{x}) = \left(\mathcal{C}_{l^*}(\mathbf{x}) \cdot \prod_{l \neq l^*} \mathcal{C}_l(\mathbf{x})\right)^{1/L} < 1$.
3. **Strict Convergence:** If $\mathbf{x}$ exhibits an "Absolute Violation" in any layer ($\mathcal{P}_{l^*} \to 1 \implies \mathcal{C}_{l^*} \to 0$), then $H(\mathbf{x}) \to 0$ regardless of the performance in other layers.

Under the assumptions above, this confirms that $H(\mathbf{x})$ behaves as a conservative diagnostic for joint satisfaction of latent manifold constraints.

A.2 Proof of Asymptotic Statistical Convergence

Theorem 2 (Asymptotic Convergence of the HIF Penalty). Let $\hat{P}_N(x_h \mid \mathbf{x}_{\setminus h})$ be the empirical conditional probability estimator (the Sentinel f_h) trained on a ground-truth dataset $\mathcal{D}_{real}$ of size N. Assuming the base estimator is statistically consistent (e.g., consistent non-parametric trees), as $N \to \infty$, $\hat{P}_N$ converges in probability to the true conditional distribution $P^*(x_h \mid \mathbf{x}_{\setminus h})$. Consequently, the empirical penalty $\mathcal{P}_h^{(N)}(\mathbf{x})$ converges pointwise in probability to the corresponding population penalty under the same modeling assumptions:

$$\lim_{N\to\infty} P\left(\left|\mathcal{P}_h^{(N)}(\mathbf{x}) - \mathcal{P}_h^*(\mathbf{x})\right| > \epsilon\right) = 0 \quad \forall \epsilon > 0,\ \mathbf{x} \in \mathcal{M}_{real} \tag{12}$$

1. **Consistent Density Estimation:** Because the LSE leverages consistent non-parametric estimators (Random Forests), the empirical conditional probabilities converge pointwise in probability to the true Bayesian posterior P^*, provided the feature space is bounded and the leaf nodes grow sub-linearly with N; the pointwise (rather than uniform) statement reflects that sup-norm convergence is not generally guaranteed for forest-based estimators.
2. **Bounded False Positive Rate:** By defining the confidence floor δ_h as the empirical 5th-percentile of $\hat{P}_N$ on the training data, we obtain an asymptotic upper bound of approximately $\alpha = 0.05$ on false positives under calibration assumptions.
3. **Asymptotic Soundness:** Therefore, synthetic records penalized by the converged HIF ($\mathcal{P}^* > 0$) are expected to lie in lower-probability conditional regions of the data manifold, supporting (but not by itself proving) a structural-violation interpretation.
4. **Consistency of Feature Dependency Score $S(h)$:** By the Uniform Law of Large Numbers (ULLN) and the Continuous Mapping Theorem, the empirical feature dependency score $S^{(N)}(h)$ converges in probability to the Bayes optimal classification accuracy $S^*(h) = \mathbb{E}_{\mathbf{x}\sim P_{real}}\left[\mathbb{I}\left(\arg\max_c P^*(x_h = c \mid \mathbf{x}_{\setminus h}) = x_h\right)\right]$, implying asymptotically stable hub ranking under these conditions.

A.3 Theoretical Bound on Type I Error and Rejection Filtering Purity

Theorem 3 (Type I Error Bound & Cohort-Purity Approximation). Let $\mathbf{x} \sim P_{real}$ be a valid record residing on the ground-truth manifold $\mathcal{M}_{real}$ (including rare tail outliers or low-density components of a Gaussian Mixture Model). Under quantile calibration ($\delta_h = Q_\alpha$ and continuous tolerance bandwidth $\gamma_y = Q_{1-\beta}$), the false-rejection probability is upper-bounded by the Union Bound:

$$P\left(H(\mathbf{x}) < 0.5 \mid \mathbf{x} \in \mathcal{M}_{real}\right) \leq \sum_{h\in\mathcal{H}} \alpha_h + \sum_y \beta_y \tag{13}$$

Furthermore, let a generative model produce a noisy mixture distribution $P_{syn} = (1-\eta)P_{real} + \eta P_{hallucinated}$, where $\eta \in [0,1]$ represents the structural hallucination rate. Rejection filtering with threshold $H \geq 0.5$ yields an approximate retained-cohort purity expression under the mixture model assumptions:

$$\text{Purity}(H \geq 0.5) = \frac{(1-\eta)(1-\alpha)}{(1-\eta)(1-\alpha) + \eta\beta_{hallucination}} \tag{14}$$

where $\beta_{hallucination}$ is the leakage probability of hallucinated records past the filter. For generative models with high hallucination rates ($\eta \approx 0.66$), the pre-filter cohort purity is $(1-\eta) = 34\%$; with a calibrated leakage rate of $\beta_{hallucination} \approx 0.02$ and $\alpha = 0.05$, filtering elevates purity to over 96% in this illustrative model, theoretically accounting for the downstream predictive utility improvements observed in downstream tasks.

B Logical Sentinel Ensemble (LSE) Implementation Details

The LSE architecture prioritizes categorical dependency modeling by identifying and enforcing high-dependency hubs across categorical features.

B.1 Feature Dependency Hub Discovery

We ranked features based on their **Feature Dependency Score** $S(h)$, which measures the classification accuracy of predicting a feature within the full dataset context (including binned numeric features). In our experiments on the U.S. Census ACS dataset, the LSE identified the following top-5 hubs: `household_income`, `housing_cost`, `poverty_status`, `education`, and `tenure`.

B.2 Confidence Floor Calibration

The confidence floor δ_h is defined as the 5th-percentile of the probability mass that a Sentinel f_h assigns to the correct category in the ground-truth data $\mathcal{D}_{real}$, with a safety floor of 0.01:

$$\delta_h = \max(\text{Percentile}_5\left(\{f_h(\mathbf{x}_h \mid \mathbf{x}_{\setminus h}) \mid \mathbf{x} \in \mathcal{D}_{real}\}\right), 0.01) \tag{15}$$

This robust approach avoids the instability of 1st-percentile thresholds in sparse data distributions and ensures sentinels remain sensitive to conditional dependency violations [6, 25].

C Extended Results and Sensitivity Analysis

C.1 Hyperparameter Sensitivity

We evaluated HIF robustness across three key hyperparameters on the Census ACS dataset (2,000 records with 5% injected violations). All experiments used $N = 5$ random seeds; we report mean $\pm$ SEM. The continuous HIF score is the primary metric; the violation rate is derived by applying the violation threshold to row-level penalties.

C.1.1 Hub Count K

Table 6 shows that the HIF score is stable across hub counts, varying by less than 1% across the swept range (0.979 at $K = 1$ to 0.987 at $K \geq 10$), suggesting that the framework's integrity signal is not dependent on a specific number of dependency hubs. The geometric mean aggregation ensures that adding more sentinels does not dilute the score.

Table 6 Sensitivity to hub count K (percentile=5, threshold=0.5).

K	HIF Score (±SEM)	Violation Rate (±SEM)
1	0.9786 ± 0.0003	0.0039 ± 0.0003
3	0.9817 ± 0.0002	0.0036 ± 0.0001
5	0.9841 ± 0.0002	0.0037 ± 0.0001
10	0.9866 ± 0.0007	0.0106 ± 0.0010
15	0.9866 ± 0.0007	0.0106 ± 0.0010

C.1.2 Confidence Floor Percentile

Table 7 shows that the HIF score is stable for percentiles 1–5, with a minor 0.3% drop at the 10th percentile. The 5th-percentile default provides the optimal balance between sensitivity to genuine violations and robustness against borderline records.

Table 7 Sensitivity to confidence floor percentile ($K = 5$, threshold=0.5).

Percentile	HIF Score (±SEM)	Violation Rate (±SEM)
1	0.9848 ± 0.0002	0.0037 ± 0.0001
3	0.9846 ± 0.0002	0.0037 ± 0.0001
5	0.9841 ± 0.0002	0.0037 ± 0.0001
10	0.9815 ± 0.0002	0.0048 ± 0.0002

C.1.3 Violation Threshold

Table 8 demonstrates that the continuous HIF score is invariant to the binary classification threshold, as expected. The violation rate scales monotonically with the threshold, allowing practitioners to tune the operating point for their specific precision–recall requirements without retraining.

C.2 Scalability and Baseline Comparisons

We benchmarked HIF against the **SHAP Distance** semantic fidelity metric [9], which serves as a state-of-the-art baseline for explainability-aware evaluation. Although the

Table 8 Sensitivity to violation threshold ($K = 5$, percentile=5).

Threshold	HIF Score (±SEM)	Violation Rate (±SEM)
0.3	0.9841 ± 0.0002	0.0081 ± 0.0002
0.5	0.9841 ± 0.0002	0.0037 ± 0.0001
0.7	0.9841 ± 0.0002	0.0021 ± 0.0001

SHAP Distance effectively identifies semantic drift, it suffers from severe scalability bottlenecks. HIF auditing scales linearly, $O(N \cdot K)$, where K is the number of hubs. In our benchmark (`scripts/11_scalability_benchmark.py`), fitting the 5-hub LSE auditor on 2,000 real records took 9.7 s, after which scoring 10,000 synthetic records took 1.50±0.15 s and scoring 100,000 records took 10.6±0.5 s on commodity multi-core hardware (fit amortized once; wall-clock averaged over three repeats). The marginal per-10,000-row cost at the 100,000-row scale (10.6 / 10 ≈ 1.1 s per 10k rows) confirms the linear scaling. In contrast, SHAP-based methods require $O(N \cdot D \cdot 2^D)$ operations or expensive approximations, which precludes exact evaluation at this volume. Thus, HIF provides comparable detection accuracy while maintaining the necessary computational efficiency to scale to high-cardinality datasets with millions of rows of data.

C.3 Privacy-Fidelity Trade-off (TAMIS)

Using the TAMIS Privacy Oracle, we audited membership-inference risk on Census ACS CTGAN cohorts ($N = 3$ seeds). The HIF Oracle retained 66.7% of synthetic records, and the MIA AUC remained at chance level relative to the full cohort (0.499 ± 0.024 vs. 0.490 ± 0.015, both consistent with random guessing), with zero exact copies detected across all variants. HIF filtering is therefore privacy-neutral under the TAMIS audit: removing logically inconsistent rows does not increase membership-inference vulnerability.

C.4 Cross-Architecture Violation Frontiers

Our Cross-Architecture Maturity Audit revealed distinct "Violation Frontiers" for different generative families. While statistical models (**Gaussian and Vine Copulas**) tend to violate constraints through excessive smoothing (averaging out manifold sharp edges), neural models like **CTGAN** exhibit *Combinatorial Violations.*

Specifically, CTGAN often fragments the latent manifold during adversarial training, leading to synthetic records that satisfy the marginal distributions for each feature but inhabit "Impossible Junctions"—categorical combinations that have zero probability mass in the ground-truth manifold. The Logical Sentinel Ensemble (LSE) is particularly effective at identifying these junctions, as the sentinels assign near-zero probability to the violated categories conditioned on their manifold hubs. This architectural discrepancy explains the higher base integrity score of CTGAN ($H = 0.285$) compared to that of Vine Copulas ($H = 0.278$) on complex relational datasets (e.g., Online Purchases), despite the latter's superior statistical marginal fidelity.

C.5 Macro-Economic Violation Detection

The application of HIF to macroeconomic domains is a natural extension: macro-level tables are governed by tight structural relationships (e.g., between GDP growth, inflation, and the Gini index), and sparse state-level semantics are a plausible failure regime for generative models. We leave a systematic evaluation on such data (e.g., World Bank Development Indicators) to future work; the results reported in this paper are confined to the microdata benchmarks for which the full pipeline and evidence are available.

D Limitations and Open Challenges

While the Hybrid Integrity Framework (HIF) provides a technically rigorous diagnostic for synthetic data, our current formulation faces three key challenges identified through empirical auditing.

D.1 Dependency Hub Selection via Feature Predictability

The Logical Sentinel Ensemble (LSE) relies on single-variable classification accuracy ($S(h)$) to discover dependency hubs. Although computationally efficient, it primarily captures categorical dependencies. Future work will investigate **Total Correlation** and multi-way interaction kernels to generalize hub selection to complex parity regimes.

D.2 Sensitivity on Sparse or Imbalanced Manifolds

The use of the **5th-percentile Confidence Floor** (δ_h) is a data-driven heuristic designed for stability, drawing on classical principles of robust percentile dispersion thresholding [47]. However, on extremely sparse or highly imbalanced datasets (e.g., rare fraud events or sparse healthcare codes), the calibration of δ_h may be susceptible to outliers in ground-truth data. In such cases, the framework may require domain-specific tuning of the percentile parameter or the adoption of Laplacian smoothing to stabilize the boundary conditions.

The parallelizable nature of the LSE architecture allows it to scale effectively to tables with thousands of columns. However, the memory overhead associated with training and storing an ensemble of K Random Forest sentinels for ultra-high-dimensional tables (e.g., $D > 10,000$) remains a potential hardware bottleneck. In such regimes, we propose replacing complex forest-based sentinels with sparse-linear or neuro-inspired models that offer a better trade-off between semantic depth and memory usage.